POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Data Communication (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the five essential components of a data communication system. How does each component contribute to successful communication? Explain with suitable diagram. 7
   b) What is serial and parallel data transmission? Differentiate between synchronous and asynchronous data transmission. 8
2. a) Check whether the given system is linear, time invariant and causal or not. 7
      i. $y(t) = \log x(t)$
      ii. $y(t) = t^2 x(t)$
   b) Differentiate between energy and power signal. Calculate whether ramp signal is power signal or energy. 8
3. a) Why Frame Relay is faster than X.25? Explain TCP/IP layers and compare it with OSI reference model. 8
   b) What is the purpose of cladding in an optical fibre? Explain the modes of propagation used in the optical fibre. 7
4. a) Define transmission impairments. And explain the causes of transmission impairments in details. 7
   b) If the 7 bit Hamming codeword received by a receiver is 1011011. Assuming the even parity, state whether the received codeword is correct or wrong. If wrong, decode the correct data word and locate the bit in error. 8

   OR

   Compress "Engineering" using Huffman codes and find its efficiency and redundancy.
5. a) What is ARQ? What are the two types of sliding window ARQ error control? Explain. 8

   b) What do you mean by multiplexing? How is TDM different from FDM? 7
6. a) Why do we need modulation need in data communication? Explain. 7

   OR

   Calculate the rate of a 500 baud 8-QAM signal with a constellation diagram.
   b) Represent the given bit stream 10011100001101 using 8
      i. Polar NRZ
      ii. Differential Manchester
      iii. HDB3
7. Write short notes on: **(Any two)** 2×5
   a) Standards organizations
   b) Noise and its types
   c) HDLC protocol vs Point-to-Point protocol

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np